TTISA-32 — Time-Travel Instruction Set Arch

Instruction	31:25	24:20	19:15	14:12	11:7
CHECKPOINT	0000000	00000	00000	000	00000
SNAPSHOT	0000000	00000	00000	001	rd
RESTORE	0000000	00000	rs1	010	00000
REWIND	0000000	00000	00000	011	00000
FORWARD	0000000	00000	00000	100	00000

TTISA-32 Instruction Encoding Format

Bits	31–25	24–20	19–15	14–12	11–7
Field	funct7	rs2	rs1	funct3	rd
Width	7 bits	5 bits	5 bits	3 bits	5 bits
TT-Type	0000000	00000	rs1/00000	TT-specific	rd/00000

32-bit instruction layout

31	30	29	28	27	26
funct7	funct7	funct7	funct7	funct7	funct7

Execution example

Step	Instruction	Timeline Position	Result

ıitecture

6:0	Format	Assembly Synta	Description
0001011	TT-Type	CHECKPOINT	Create a checkpoint of the current architectural
0001011	TT-Type	SNAPSHOT rd	Save the current state and write the snapshot
0001011	TT-Type	RESTORE rs1	Restore the checkpoint identified by the snapshot
0001011	TT-Type	REWIND	Move to the previous available checkpoint
0001011	TT-Type	FORWARD	Move to the next available checkpoint

6–0	Meaning
opcode	Instruction fields
7 bits	Total = 32 bits
0001011	Custom time-travel instruction format

25	24	23	22	21	20	19	18	17	16	15	14	13	12	11
funct7	rs2	rs2	rs2	rs2	rs2	rs1	rs1	rs1	rs1	rs1	funct3	funct3	funct3	rd

10	9	8	7	6	5	4	3	2	1	0
rd	rd	rd	rd	opc ode	opc ode	opc ode	opc ode	opc ode	opc ode	opc ode

1	CHECKPOINT	CP0	State is saved		
2	ADD R3,R1,R2	S1	Normal execution		
3	CHECKPOINT	CP1	Second state is saved		
4	SUB R4,R3,R1	S2	Normal		
5	REWIND	CP1	CPU returns		
6	ADD R5,R1,R2	S1’	New		
